

SOP-06: Ethidium Bromide (PHS)

Roberts Lab, School of Aquatic and Fishery Sciences - Rev. 1 - September 11, 2026

PARTICULARLY HAZARDOUS SUBSTANCE (PHS): YES - Sections 9, 10 and 11 are MANDATORY and are completed below.

Section 1 - Lab-Specific Information

Building/Room(s) covered by this SOP	Fisheries Teaching & Research (FTR) Building, Room 209
Unit or department	School of Aquatic and Fishery Sciences (SAFS), College of the Environment
Principal Investigator	Steven Roberts (work phone 206-685-3742)
PI signature / date	_____ Date: _____
This SOP was created by (if not PI)	_____ Date: _____
Emergency contact	9-1-1 (campus phone) EH&S 206-543-7262 (M-F 8-5) 206-685-UWPD (8973) after hours
Date of last review	_____ Next review due: _____ (review annually and after any incident or process change)

Chemicals covered by this SOP

Chemical name (as listed in MyChem)	CAS number
Ethidium bromide, solid (1 g, catalog 4007)	1239-45-8
Ethidium bromide working solutions (10 mg/mL stock; 0.1 microgram/mL final in stained agarose gels only - not added to running buffer)	1239-45-8

Safety Data Sheets for every chemical listed above must be readily available in the laboratory. SDSs are accessible through MyChem.

Section 2 - Hazards

Applicable GHS pictograms: Skull and crossbones (acute toxicity Cat 2, inhalation); Health hazard (germ cell mutagenicity Cat 2); Exclamation mark

- PARTICULARLY HAZARDOUS SUBSTANCE. Ethidium bromide is a potent frameshift mutagen (GHS Germ cell mutagenicity Category 2) and is highly toxic by inhalation (GHS Acute toxicity Category 2, inhalation). It intercalates into DNA and is regarded as a suspect carcinogen and reproductive hazard even though it lacks a formal carcinogen classification. Sections 9-11 of this SOP are mandatory.
- Stock chemicals: 1 g of ethidium bromide POWDER. Intermediates and working solutions: 10 mg/mL aqueous stock, added directly to molten agarose to cast stained gels at 0.1 microgram/mL final (1 uL

of 10 mg/mL stock per 100 mL of gel volume). Ethidium bromide is NOT added to the 1X TAE/TBE electrophoresis (running) buffer - only the gel itself is stained. Wastes: stained gels, contaminated gloves, tips and wipes, and any unused powder.

- The powder form is by far the greatest hazard in this inventory. Weighing ethidium bromide powder creates a respirable, highly toxic, mutagenic dust. Wherever possible, purchase and use a pre-made 10 mg/mL solution or a non-mutagenic alternative stain (SYBR Safe, GelRed, GelGreen) instead of weighing powder. This SOP treats weighing of the powder as a restricted operation requiring specific PI authorization.
- Routes of exposure: inhalation of powder or aerosol (primary hazard, acute toxicity Category 2), skin contact and absorption, eye contact, ingestion.
- How exposure might occur: weighing or transferring the dry powder; preparing the 10 mg/mL stock; adding stain to molten agarose; pouring, handling, photographing and disposing of stained gels; contact with contaminated gel boxes, buffer, pipettes and the UV transilluminator surface; and cross-contamination from gloves to door handles, keyboards and pens.
- Target organs: genetic material of all dividing cells; also skin, eyes and respiratory tract. Because the compound is mutagenic, there is no identified threshold below which exposure is considered risk-free; exposure is minimized as far as practicable.
- Signs and symptoms of exposure: skin, eye and upper respiratory irritation. Mutagenic effects produce no acute signs, which is precisely why strict work practice controls, not symptom-watching, are the basis of protection.
- Note that ethidium bromide work is normally accompanied by UV transilluminator use. UV-B/UV-C radiation causes photokeratitis and skin burns; a UV-blocking face shield and closed clothing are required whenever the transilluminator is on, and a blue-light or gel-doc enclosure is preferred.
- GHS classification (from MyChem / SDS): Acute toxicity Category 2 (inhalation); Acute toxicity Category 4 (oral); Germ cell mutagenicity Category 2. MyChem fire class: Highly Toxic; Other Health Hazards - Including Carcinogens; Irritant.
- There is no established OSHA PEL, WISHA PEL or ACGIH TLV for ethidium bromide. Control is based on eliminating exposure rather than on an exposure limit.

Section 3 - Engineering Controls and Personal Protective Equipment (PPE)

Engineering controls

- Weighing or any handling of ethidium bromide POWDER is performed ONLY inside the certified chemical fume hood in FTR 209, on a disposable absorbent liner, using a balance placed inside the hood, with the sash at or below the marked working height. Any chemical fume hood used must be tested and passed by EH&S. Weighing powder on an open bench balance is prohibited.
- Preparation of the 10 mg/mL aqueous stock from powder is performed in the fume hood. Once in solution, the inhalation hazard is greatly reduced.
- Addition of ethidium bromide stock to molten agarose and pouring of gels is performed at the designated gel bench in FTR 209 on a disposable absorbent bench liner, with the molten agarose at or below 60 degC to limit aerosol.
- Electrophoresis, gel handling and imaging are performed at the designated gel bench and gel-doc station only. A UV-blocking face shield is used with any open UV transilluminator; an enclosed gel documentation system is preferred and is used when available.

- The designated gel area is physically separated from general bench work, is covered with a disposable absorbent liner, and holds its own dedicated set of pipettes, tips, gloves, marker, and waste containers that are not used elsewhere.
- The nearest eyewash and safety shower are located in FTR 209 and are inspected and flushed weekly by lab staff.

Hygiene measures

- Avoid contact with skin, eyes, and clothing. Wash hands after removing PPE, before breaks, and immediately after handling the chemical. If any chemical covered by this SOP comes into contact with any PPE, the PPE shall be immediately removed and discarded properly. Any potentially exposed body parts should be washed immediately.
- No eating, drinking, chewing gum, applying cosmetics, or storing food or drink in any area where these chemicals are used or stored.

Skin and body protection

- Chemically compatible laboratory coats that fully extend to the wrist must be worn, appropriately sized and buttoned to their full length. Personnel must also wear full-length pants, or equivalent, and close-toe shoes. The area of skin between the shoe and ankle must not be exposed.
- Lab coat type: buttoned barrier lab coat, dedicated where practical to the gel area. A disposable sleeve set is worn when weighing powder or preparing the 10 mg/mL stock.
- Because ethidium bromide is toxic by skin absorption and is a mutagen, additional protective clothing (disposable sleeves; apron for stock preparation) is required where splashes or skin contact are foreseeable. A UV-blocking face shield is worn whenever an open transilluminator is on.
- Contaminated lab coats are removed immediately, bagged and handled as contaminated waste or decontaminated; they are never taken home.

Hand protection

- Hand protection IS required for the activities described in this SOP.
- Required gloves: DOUBLE 4-6 mil nitrile gloves for all work with the powder and the 10 mg/mL stock. Single 4-6 mil nitrile gloves are acceptable for handling stained gels and running buffer, changed immediately on any visible contact and at the end of each task.
- Latex gloves must NOT be used - ethidium bromide penetrates latex readily.
- Consult your preferred glove manufacturer to ensure that the gloves you plan to use are compatible with ethidium bromide. Gloves must be inspected prior to use, including a check for pinholes.
- Gloves worn at the gel bench must NEVER be worn elsewhere. Remove gloves before touching door handles, keyboards, phones, pens, refrigerator handles or your face. This is the most common route by which ethidium bromide contaminates a laboratory.
- Use proper glove removal technique (without touching the glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices. Wash and dry hands immediately after glove removal.

Eye protection

- ANSI Z87.1-compliant eye protection IS required for all work with ethidium bromide. Ordinary prescription glasses will NOT provide adequate protection unless they also meet the Z87.1 standard and have compliant side shields.

- Minimum eye protection: ANSI Z87.1 safety glasses with side shields for routine gel casting, loading and handling; chemical splash goggles for weighing powder and preparing or diluting the 10 mg/mL stock.
- A UV-blocking full-face shield (in addition to safety glasses) is REQUIRED whenever an open UV transilluminator is energized.

Respiratory protection

- Respiratory protection is NOT required for the activities described in this SOP, provided all powder handling is confined to the certified chemical fume hood as written.
- Ethidium bromide POWDER is never permitted to be handled outside of the chemical fume hood. The stained gel (0.1 microgram/mL) and the 10 mg/mL stock, once made, may be used at the designated gel bench outside the fume hood. The electrophoresis running buffer is not stained under this SOP.
- Respiratory protection may be needed if a dust or aerosol hazard is present and work is conducted outside of the fume hood; because such work is prohibited here, no respirator is needed. If any procedure may pose such a hazard, it should be eliminated or strictly isolated.
- Respirators should be used as a last line of defense (i.e., after engineering and administrative controls have been exhausted). If a potential exposure hazard cannot be eliminated, contact the EH&S Respiratory Protection Program administrator to discuss respiratory protection or to enroll in the program; enrollment includes medical evaluation, training and fit testing.

Section 4 - Special Handling and Storage Requirements

- Substitution first: before any new use of ethidium bromide, the user must consider a non-mutagenic alternative stain (SYBR Safe, GelRed, GelGreen). The PI must approve continued ethidium bromide use for each project.
- Quantity limits: no more than 100 mg of powder is weighed in a single operation, and only enough 10 mg/mL stock is prepared for approximately one year of use. Working gel concentration is 0.1 microgram/mL (1 μ L of 10 mg/mL stock per 100 mL of gel volume). Ethidium bromide is added only to the molten agarose, never to the electrophoresis running buffer.
- The powder container is opened only in the fume hood and is kept in a labeled, closed secondary container at all other times.
- All containers - powder bottle, stock tube, waste containers - must be labeled with the full chemical name, concentration, the words 'MUTAGEN - PARTICULARLY HAZARDOUS SUBSTANCE', the date and the preparer's initials. The electrophoresis running buffer is not stained under this SOP and does not require this labeling unless it becomes visibly contaminated (see Section 6).
- A dedicated set of pipettes, tip boxes, a marker, a gel box and a gel tray are reserved for ethidium bromide work and are labeled 'EtBr ONLY'. They are not used for any other purpose and are not removed from the designated area.
- Monitoring for spread: the designated gel area, the transilluminator surface, adjacent bench edges, pipette barrels and the nearest door handle are checked at least quarterly for fluorescent contamination using a hand-held UV lamp in a darkened room, and the check is documented. Any fluorescence found is decontaminated and the survey repeated.
- Unattended operations: gels may be run unattended only inside the designated area with the lid on the gel box, a posted sign giving the chemical, the responsible person and a contact number, and the power supply on a timer or under a documented check-in.

- Storage: ethidium bromide powder and the 10 mg/mL stock are stored in a labeled, closed, opaque secondary container in the dedicated toxics cabinet in FTR 209, protected from light, below eye level, segregated from oxidizers and strong acids. The secondary container carries the PHS designation sign.
- Transport: the powder bottle is not transported outside FTR 209. Solutions are carried in closed, leak-proof secondary containers.
- Clean the chemical fume hood upon completion of powder tasks with a detergent solution followed by clean water; wipe dry, and survey with a UV lamp.
- Clean all contaminated surfaces at the gel bench with detergent solution and dry; do not use bleach (see Section 6).
- Place all contaminated disposable items in the dedicated ethidium bromide solid waste container.
- Non-disposable/re-usable utensils, gel boxes, trays, combs and other surfaces contaminated with ethidium bromide must be decontaminated at the end of the laboratory work session, within the designated area, before removing any of the items.
- When work is completed, remove gloves and wash hands with soap and water.

Section 5 - Spill and Accident Procedures

Spill response

- Chemical spills must be cleaned up as soon as possible by properly protected and trained personnel. All other persons should leave the area. Do not attempt to clean up any spill if not trained or comfortable. Evacuate the area and call 9-1-1 on a campus phone for help.
- POWDER spill of any size: this is the most serious credible incident under this SOP because the material is acutely toxic by inhalation. If the spill is contained inside the operating fume hood and is less than approximately 100 mg, trained staff may clean it up: leave the hood running, do NOT brush or wipe dry, dampen the powder with water using a wash bottle, then wipe up with damp absorbent working from the outside inward, and place all material in the dedicated solid waste container. A powder spill OUTSIDE the fume hood, or larger than approximately 100 mg: evacuate the room, close the door, post a warning sign, prevent entry, and call 9-1-1. Notify EH&S.
- LIQUID spill - less than 100 mL of molten stained agarose (0.1 microgram/mL) during casting at the designated bench: may be handled by trained lab staff wearing safety glasses, double nitrile gloves and a lab coat. Absorb with paper towels or absorbent pads, working from the outside inward, place in the dedicated solid waste container, then wash the area with detergent solution and re-survey with a UV lamp until no fluorescence remains.
- LIQUID spill of the 10 mg/mL stock, or any liquid spill greater than 100 mL, or any spill outside the designated area: restrict access, post a sign, and contact EH&S at 206-543-0467 during business hours or via 9-1-1 after hours.
- Do NOT use bleach to decontaminate ethidium bromide. Hypochlorite degrades ethidium bromide into products that have been reported to be mutagenic, and it complicates waste disposal. Use detergent and water and dispose of the material as hazardous waste.
- PPE required for cleanup: chemical splash goggles, double nitrile gloves, buttoned lab coat with disposable sleeves, full-length pants and close-toe shoes; add a face shield for stock-solution or powder spills.
- Signage and entry restrictions: post 'DO NOT ENTER - MUTAGEN SPILL' on the door for any powder spill outside the hood; restrict entry until EH&S clears the space.

- Spill area must be cleaned up in the following manner: wash the area thoroughly with detergent solution, followed by clean water, and dry; then survey with a hand-held UV lamp in a darkened room and repeat until no fluorescence is visible.
- Spill cleanup materials must be disposed of in the following manner: place all absorbent, wipes and gloves in the dedicated, labeled ethidium bromide solid hazardous waste container; double bag, label with a UW Hazardous Waste Label listing 'ethidium bromide - spill debris', and submit a collection request to EH&S.
- For questions on spill cleanup, contact EH&S spill consultants at 206-543-0467 during normal business hours (Monday-Friday, 8 a.m. to 5 p.m.).
- Any spill, exposure or near miss incident requires the involved person or supervisor to complete and submit an incident report on the EH&S website within 24 hours (certain types of incidents require immediate notification).

Exposure procedures - first aid

- Perform first aid immediately. Refer to the SDS for additional chemical-specific guidance.
- Inhalation exposure (powder): move the person out of the contaminated area to fresh air immediately and call 9-1-1. Ethidium bromide is acutely toxic by inhalation (Category 2); any inhalation of the powder is a medical emergency.
- Skin exposure: wash the affected area with soap and running water for at least 15 minutes; for larger contact use the nearest safety shower for 15 minutes, remove contaminated clothing while under the shower, and use a clean lab coat or spare clothing for cover-up. Seek medical evaluation.
- Eye exposure: use the eyewash for 15 minutes while holding the eyelids open; seek medical evaluation.
- Ingestion: rinse mouth with water; do not induce vomiting; call 9-1-1.
- Sharps injury (needle stick or subcutaneous exposure): scrub the exposed area thoroughly for 15 minutes using warm water and sudsing soap; seek medical evaluation.
- Get help. Call 9-1-1 or go to the nearest Emergency Department; provide details of exposure: agent (ethidium bromide - state that it is a mutagen and is highly toxic by inhalation), dose, route of exposure, and time since exposure. Bring the SDS and this SOP to the Emergency Department.
- Notify your supervisor as soon as possible for assistance. Secure the area before leaving; lock doors and indicate the spill if needed.
- Notify EH&S immediately after providing first aid and/or getting help. During business hours (M-F, 8-5) call 206-543-7262. Outside of business hours call 206-685-UWPD (8973) to be routed to EH&S staff on call.
- Report every exposure to ethidium bromide to the PI and EH&S, however minor it appears, because the effects of a mutagen are not detectable by symptom.

Section 6 - Waste Accumulation and Disposal Procedures

- Waste streams: (a) unused ethidium bromide powder; (b) 10 mg/mL stock solution; (c) stained agarose gels; (d) contaminated solid debris (gloves, tips, wipes, absorbent liners, weigh boats). The electrophoresis running buffer is not stained under this SOP and is not a routine EtBr waste stream - see the note below.
- Ethidium bromide is added only to the molten agarose under this SOP; the electrophoresis running buffer in the tank is not intentionally stained and is disposed of or reused as ordinary 1X TAE/TBE buffer, not as EtBr-contaminated waste. Stained agarose gels at the 0.1 microgram/mL working

concentration used here are approximately 0.00001% (1×10^{-5} %) w/v ethidium bromide - far below the 0.1% w/v threshold below which UW EH&S (<https://www.ehs.washington.edu/chemical/chemical-treatment-and-recycling>) treats stained gels as non-hazardous, so gels are trash-eligible as described below. If running buffer or rinse water is ever visibly pink or red (indicating gel dye has bled into it, or a spill), treat it as EtBr-contaminated liquid: solutions confirmed below 10 micrograms per liter (10 ppb) may be disposed of down the drain; anything at or above that threshold, or not confirmed, requires the carbon-adsorption treatment in the next paragraph before drain disposal, or collection as hazardous waste. The unused powder and the 10 mg/mL stock solution remain hazardous waste at all times and must never be poured down the drain.

- Do NOT bleach or photodegrade ethidium bromide waste in the lab. Hypochlorite treatment produces mutagenic degradation products. The only in-lab treatment authorized under this SOP is activated-carbon adsorption of EtBr-contaminated liquid (expected only if running buffer or rinse water becomes visibly discolored), as described below.

Treatment of EtBr-contaminated liquid (a visibly discolored running buffer, or gel-soak/rinse water from equipment decontamination): use activated-carbon adsorption bags (e.g., GreenBags or an equivalent EH&S-recommended product) per the manufacturer's instructions. Once the ethidium bromide has been adsorbed, the treated water may be disposed of down the drain with running water. The spent carbon bags adsorb ethidium bromide tightly enough to be discarded as regular solid waste; double-bag, label "non-hazardous - treated EtBr adsorbent," and discard in the regular trash.

- Stained agarose gels containing less than 0.1% ethidium bromide (all gels cast at the working concentration used in this SOP - see Section 6 above) may be double-bagged, labeled "non-hazardous," and discarded in the regular trash. A gel that appears pink or red, indicating a substantially higher concentration, is not covered by this exception and must be managed as hazardous waste. Pipette tips that contacted dilute ethidium bromide are air-dried and then placed in the laboratory broken-glass/sharps container, not the liquid or general solid waste stream.
- Accumulate any EtBr-contaminated liquid that does not qualify for drain disposal (the 10 mg/mL stock, and any visibly discolored running buffer or rinse water prior to carbon treatment) at the point of generation - the designated gel bench - in a sturdy, chemically compatible, opaque or foil-wrapped HDPE container with a securely closable screw-top lid, kept closed except when actively adding waste, and in secondary containment.
- Solid debris that does not qualify for the trash exceptions above - gloves, wipes and weigh boats from powder or 10 mg/mL stock handling, pink/red gels, and any absorbent used for a spill or for treated-liquid handling before it is confirmed non-hazardous - is accumulated in a dedicated, lined, closed container at the designated bench, labeled 'ETHIDIUM BROMIDE SOLID WASTE - MUTAGEN'.
- All chemical waste containers must be labeled with a UW Hazardous Waste Label listing every constituent and approximate concentration, starting from the moment the first drop or gel is added. Refer to 'How to Label Chemical Waste Containers' on the EH&S website.
- Manage chemical and hazardous chemical waste separately from other waste streams such as biohazardous waste. Never autoclave chemical waste because it can produce hazardous chemical vapors, aerosols, and explosive reactions.
- The University of Washington Environmental Health & Safety Department has full responsibility for collection of hazardous waste for the University; University laboratories cannot contract with an outside vendor to collect hazardous waste.
- To request a collection of chemical waste, submit a form on the Chemical Waste Disposal webpage on the EH&S website or directly in MyChem inventory. Contact EH&S at 206-616-5835 or chmwaste@uw.edu with questions.
- Because the 1 g powder container represents a long-term inventory hazard and a non-mutagenic substitute is available, the PI should consider surplus the remaining powder through EH&S. Update the MyChem inventory when containers are emptied, received, transferred or surplus.

- Visit the Hazardous Material Disposal and Recycling webpage on the EH&S website for information on disposing, recycling and surplusing materials.

Section 7 - Protocol

- Procedure A - Preparation of 10 mg/mL stock from powder (RESTRICTED - requires specific written PI authorization for each occasion). Inside the certified fume hood, on a disposable absorbent liner, with the balance inside the hood, wearing goggles, double nitrile gloves, lab coat and disposable sleeves: weigh the powder into a tared, capped tube using a disposable weigh boat and spatula. Add nuclease-free water inside the hood, cap, and dissolve by inversion. Wipe the tube exterior, label it 'ETHIDIUM BROMIDE 10 mg/mL - MUTAGEN', date and initial. Discard the weigh boat, spatula, liner and gloves into the dedicated solid waste container. Wipe the balance and hood with detergent solution and survey with a UV lamp.
- Procedure B - Casting stained agarose gels. At the designated gel bench, on an absorbent liner: melt agarose in TAE or TBE, cool to approximately 55-60 degC, add 1 uL of 10 mg/mL stock per 100 mL of buffer (0.1 microgram/mL final) using a dedicated pipette, swirl gently (do not create aerosol), and pour into the gel tray. Discard the tip into the dedicated solid waste container.
- Procedure C - Electrophoresis and imaging. Load and run the gel with the lid on. Wear a UV-blocking face shield over safety glasses whenever an open transilluminator is energized; use the enclosed gel-doc system in preference. Handle the gel only with gloved hands over the absorbent liner. Return the gel directly to the dedicated solid waste container after imaging.
- Procedure D - Buffer and equipment handling. The electrophoresis running buffer is not stained under this SOP and may be reused for subsequent runs or disposed of as ordinary 1X TAE/TBE buffer (drain disposal, no special labeling), unless it appears visibly pink or red, in which case treat it as EtBr-contaminated liquid per Section 6. Rinse the gel box and tray with a small volume of water at the designated bench - this rinse water follows the same rule as the running buffer. Wash with detergent and water, and survey with a UV lamp before returning equipment to the 'EtBr ONLY' storage location.
- Prohibited: weighing powder outside the fume hood; bleach treatment of any ethidium bromide material; drain disposal of the powder, the 10 mg/mL stock, or any visibly EtBr-contaminated liquid that has not been carbon-treated (or confirmed below 10 micrograms per liter); use of latex gloves; carrying gel-area gloves outside the designated area; use by untrained or unauthorized personnel.
- Refer to Section 2 of the UW Laboratory Safety Manual on the EH&S website for additional guidance on chemical management and preparation for use for particularly hazardous substances (PHSs).
- NOTE: Any deviation from this SOP requires approval from the Principal Investigator.

Section 8 - Special Precautions for Animal Use

No. N/A - ethidium bromide is not administered to animals under this SOP and is used only on extracted nucleic acids.

IACUC protocol number (if applicable): _____

PARTICULARLY HAZARDOUS SUBSTANCE (PHS): YES - Sections 9, 10 and 11 are MANDATORY and are completed below.

Section 9 - Approvals Required

- All staff working with ethidium bromide must be trained on this SOP prior to starting work. They must also review the ethidium bromide SDS, and it must be readily available in the laboratory (MyChem). All training must be documented and maintained by the PI or their designee.
- Ethidium bromide is a particularly hazardous substance. Written authorization from the Principal Investigator is required before any individual may work with it, and that authorization is recorded on the training roster in Section 12 with the PI's countersignature.
- Handling the POWDER (Procedure A) is a restricted operation requiring specific written PI authorization on each occasion, in addition to general authorization. The PI must first confirm that a pre-made solution or a non-mutagenic alternative stain cannot be used.
- Prerequisite training, all documented before independent work: (1) EH&S Laboratory Safety Training, initial and current refresher; (2) EH&S Managing Laboratory Chemical Waste training; (3) documented review of this SOP and the SDS; (4) documented instruction on UV transilluminator safety; (5) documented instruction on glove discipline and contamination surveys.
- A worker must complete at least one supervised run of Procedures B-D, observed and signed off by the PI or a designated experienced lab member, before working with ethidium bromide independently. Two supervised runs are required before Procedure A may be performed.
- Personnel who are pregnant, planning a pregnancy, or breastfeeding should consult the EH&S Reproductive Health in the Workplace resources and their healthcare provider and are encouraged to discuss alternative task assignment with the PI, because ethidium bromide is a germ cell mutagen Category 2. Such consultation is voluntary and confidential.
- No medical examination is required for the work described because respirators are not used.

Section 10 - Decontamination

- Do not use bleach. Decontamination is by detergent and water plus removal of the material as hazardous waste, verified by UV survey.
- Equipment: wash gel boxes, trays, combs and dedicated pipette barrels with laboratory detergent solution and water at the designated bench, rinsing into the ethidium bromide liquid waste container for the first rinse. Dry and survey with a hand-held UV lamp in a darkened room; repeat until no fluorescence is detectable.
- Transilluminator and gel-doc surfaces: wipe with detergent solution on a damp wipe, then clean water, then dry. Do not use ethanol alone, which spreads rather than removes the stain. Survey with a UV lamp.
- Fume hood and balance (after powder work): wipe with detergent solution, then clean water, dry, and survey with a UV lamp. Discard the absorbent liner as hazardous solid waste.
- Designated bench: replace the absorbent bench liner at the end of each session and discard it as hazardous solid waste. Wipe the underlying bench with detergent solution monthly and survey.
- Controlled areas: there are no glove boxes or restricted-access hoods in FTR 209; the designated gel area described in Section 11 is the controlled area for this SOP and is surveyed quarterly with the results documented.
- Cleaning solutions and materials: laboratory detergent (Alconox or Contrad) solution, clean water, lint-free wipes, disposable absorbent bench liners, and a hand-held long-wave UV lamp for verification.

Section 11 - Designated Area

- DESIGNATED AREA 1 - powder handling and stock preparation: the certified chemical fume hood in FTR 209, with the balance placed inside the hood and a disposable absorbent liner in place.
- DESIGNATED AREA 2 - gel casting, electrophoresis, imaging and gel-area waste: the dedicated agarose gel bench and adjacent gel-doc station in FTR 209, covered with a disposable absorbent liner and holding the dedicated 'EtBr ONLY' pipettes, tips, marker, gel boxes and trays.
- Both designated areas are posted with a 'DESIGNATED AREA - PARTICULARLY HAZARDOUS SUBSTANCE - ETHIDIUM BROMIDE - MUTAGEN - NO BLEACH' sign listing the PI, contact number, required PPE and the authorization requirement.
- Ethidium bromide liquid and solid waste containers are kept within Designated Area 2 in secondary containment.
- Stock storage: labeled, closed, opaque secondary container in the toxics cabinet in FTR 209, posted with the same PHS designation.
- No other bench, room or piece of equipment in the Roberts Lab space is authorized for ethidium bromide use. Gloves, pipettes and equipment from the designated areas do not leave them.

Section 12 - Documentation of Training

- Prior to using substances included in this SOP, laboratory personnel must be trained on the hazards described in this SOP, how to protect themselves from the hazards, and emergency procedures.
- Ready access to this SOP and to a Safety Data Sheet for each hazardous material described in the SOP must be made available in the lab space(s) where these substances are used.
- The Principal Investigator, or Responsible Party if the activity does not involve a PI, must ensure that their laboratory personnel have attended appropriate laboratory safety training (and refresher training where applicable).
- Training must be repeated following any revision to the content of this SOP.
- Training must be documented. This training sheet is provided as one option; other forms of training documentation (including electronic) are acceptable but records must be accessible and immediately available upon request.
- Because this SOP covers a particularly hazardous substance, the Principal Investigator countersigns each entry below to record the written authorization required by Section 9.

I have read and understand the content of this SOP:

Name (print)	Signature	Date

Name (print)	Signature	Date